A PROJECT REPORT

on

"An Ensemble Hybrid Approach for Plant Disease Prediction Using Biosensor and Soil Data"

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BY

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CERTIFICATE

This is to certify that the project entitled

"An Ensemble Hybrid Approach for Plant Disease Prediction Using Biosensor and Soil Data"

submitted by

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is a record of bonafide work carried out by them, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Engineering (Information Technology) at KIIT Deemed to be University, Bhubaneswar. This work is done during year 2025-2026, under our guidance.

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Project Guide

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Signature of Sree Harshitha

ABSTRACT

Plant diseases significantly impact global agricultural productivity, leading to an estimated 20–40% loss in crops each year. Conventional detection methods depend mainly on visual observation, which only identifies diseases after symptoms become visible—often when considerable damage has already occurred and preventive measures are less effective.

This project introduces an early warning system designed to assess plant health before visible symptoms develop, using continuous data from biosensors and soil sensors. A hybrid ensemble approach is implemented by combining Random Forest and Support Vector Machine (SVM) models through a stacking classifier. The dataset consists of 1,200 observations collected from 10 plants at 6-hour intervals over a period of about one month. It includes 11 environmental and soil-related features such as soil moisture, ambient temperature, humidity, soil pH, nitrogen, phosphorus, potassium, chlorophyll content, and electrochemical signals.

The proposed stacking model achieved approximately 98% prediction accuracy, outperforming the individual Random Forest and SVM models. Additionally, a Disease Risk Score (ranging from 0 to 100) was developed to measure the severity of plant health conditions beyond simple classification. SHAP-based explainability techniques were also applied to determine the contribution of each feature to the model's predictions. The findings indicate that chlorophyll content and electrochemical signals are the most influential indicators of plant stress.

Keywords: Plant Disease Prediction, Random Forest, Support Vector Machine, Hybrid Ensemble, Biosensor Data, Disease Risk Score, Precision Agriculture, SHAP Explainability, Early Detection.

Chapter 1
Introduction

Plant diseases affect crop production in a big way and lead to large losses every year. It is estimated that around 20–40% of crops are damaged due to diseases and pests. This creates serious problems for farmers, especially in countries like India where agriculture plays an important role.

Most farmers detect plant diseases by observing the plant manually. They look for visible signs like yellow leaves, spots, or wilting. The problem with this method is that the disease is noticed only after it has already affected the plant. By that time, it may have also spread to nearby plants, making it harder to control.

With the help of IoT and sensor technology, it is now possible to monitor plant conditions continuously. Sensors can measure different parameters such as temperature, humidity, soil moisture, pH, nutrients, and chlorophyll levels. These values give useful information about plant health even before visible symptoms appear.

Machine learning can be used to analyze this sensor data and find patterns related to plant stress or disease. By training models on this data, it becomes possible to predict the health status of plants at an early stage.

In this project, a hybrid model is developed by combining Random Forest and Support Vector Machine (SVM) using a stacking method. This model uses biosensor and soil data to predict plant health before symptoms appear.

The main contributions include a hybrid RF-SVM stacking model achieving around 98% accuracy, a Disease Risk Score (0–100) to represent severity level, SHAP-based explanation to understand feature importance, use of chlorophyll content and electrochemical signal as key features, and comparison of Random Forest, SVM, and hybrid model performance.

The main contributions are:

- A hybrid RF-SVM stacking ensemble model achieving ~98% prediction confidence
- A continuous Disease Risk Score (0-100) quantifying disease severity beyond class labels
- SHAP-based explainability identifying which sensor triggers each prediction
- Use of Electrochemical Signal and Chlorophyll Content as novel predictive features
- Three-way model comparison: Random Forest vs SVM vs Hybrid Stacking

The remainder of this report is structured as follows: Chapter 2 presents the literature review and basic concepts. Chapter 3 defines the problem statement and system design. Chapter 4 covers implementation, results, and analysis. Chapter 5 describes standards adopted. Chapter 6 concludes with future scope.

Chapter 2
Basic Concepts / Literature Review

2.1 Machine Learning for Plant Disease Detection

Machine learning has emerged as a powerful tool for agricultural applications. Supervised classification algorithms learn patterns from labeled training data and generalize to unseen samples. For plant disease prediction, the input features are sensor readings and the output labels are health status categories.

2.2 Random Forest

Random Forest, introduced by Breiman (2001), is an ensemble learning method that constructs multiple decision trees during training and outputs the class that is the mode of the classes (classification) of the individual trees. Each tree is built on a bootstrap sample of the training data, and at each split, a random subset of features is considered. This randomization reduces overfitting and improves generalization. Random Forest provides feature importance scores, making it highly interpretable for agricultural applications.

2.3 Support Vector Machine (SVM)

Support Vector Machine, proposed by Cortes and Vapnik (1995), is a supervised learning algorithm that finds the optimal hyperplane separating classes by maximizing the margin between the nearest data points (support vectors) of each class. The kernel trick — particularly the Radial Basis Function (RBF) kernel — enables SVM to handle non-linear decision boundaries by implicitly mapping data to higher-dimensional spaces. SVM is particularly effective on scaled, structured sensor data.

2.4 Hybrid Ensemble / Stacking

Stacking (stacked generalization) is an ensemble technique where predictions from multiple base models (Random Forest and SVM in this project) are used as input features to a meta-learner (Logistic Regression). This allows the meta-learner to learn how to best combine the base models' predictions, typically achieving higher accuracy than any individual model. The approach was formalized by Wolpert (1992).

2.5 Literature Review

Mohanty et al. (2016) applied deep convolutional neural networks (CNN) to the PlantVillage dataset of 54,306 leaf images and achieved 99.35% accuracy. However, this

approach requires high-quality cameras, significant compute resources, and can only detect disease after visible leaf symptoms appear — making it impractical for field-scale pre-symptomatic detection.

Ferentinos (2018) further explored CNN architectures for plant disease recognition via leaf images, confirming high accuracy but noting the same limitation: image-based models are inherently reactive rather than predictive.

Pantazi et al. (2019) proposed supervised machine learning using soil and crop sensor data for disease and stress detection, demonstrating that Random Forest achieves strong performance on structured tabular data from agricultural sensors.

Ahmed et al. (2021) compared SVM and Random Forest on datasets containing environmental parameters such as temperature, humidity, and soil nutrients. Both models achieved over 90% accuracy, with Random Forest outperforming SVM in multi-class scenarios.

Unlike all the above approaches, the proposed work uses a hybrid ensemble model on biosensor data including Electrochemical Signal and Chlorophyll Content — enabling automated, pre-symptomatic, real-time plant health prediction without requiring any visual inspection or imaging.

2.6 Key Tools and Technologies

This project was developed using Python. Scikit-learn was used for building machine learning models like Random Forest, SVM, and stacking. Pandas and NumPy were used for handling data. Matplotlib and Seaborn were used for visualization. Joblib was used to save models, and Jupyter Notebook was used for coding and testing.

Chapter 3
Problem Statement / Requirement Specifications

Problem Statement / Requirement Specifications

The main objective of this project is to predict the health status of plants using biosensor and soil sensor data before visible symptoms appear. The system should classify plants into three categories: Healthy, Moderate Stress, and High Stress. In addition to classification, the system should also generate a Disease Risk Score ranging from 0 to 100 to indicate the severity of plant stress.

3.1 Project Planning

The project was carried out in several stages. First, the dataset was collected and explored to understand its structure and check for missing values. Next, preprocessing was done by removing unnecessary columns, encoding labels, scaling features, and splitting the data into training and testing sets. After that, exploratory data analysis was performed using visualizations. Then, machine learning models such as Random Forest and SVM were trained, followed by a hybrid stacking model. The models were evaluated using different performance metrics, and finally, the trained model was used to make predictions on new data.

3.2 Project Analysis

The dataset contains 1,200 sensor readings collected from 10 plants over a period of about one month, with readings taken every 6 hours. After removing columns like Timestamp and Plant_ID, 11 numerical features were used for prediction. The target variable represents plant health status and includes three classes: Healthy, Moderate Stress, and High Stress. The dataset was found to have no missing values and had a balanced distribution of classes, making it suitable for training machine learning models.

3.3 System Design

3.3.1 Design Constraints

The system was developed using basic hardware and software resources. A standard laptop or desktop with Python support was sufficient, and no GPU was required. The software tools used include Python along with libraries such as Scikit-learn, Pandas,

NumPy, Matplotlib, and Seaborn. The dataset used is in CSV format and contains biosensor and soil sensor readings collected from plants.

3.3.2 System Architecture

The system follows a sequential machine learning pipeline:

• Step 1: Raw sensor data in CSV format is loaded using Pandas
• Step 2: Preprocessing is performed by dropping unnecessary columns, encoding labels, applying StandardScaler, and splitting the data into 80% training and 20% testing
• Step 3: Exploratory Data Analysis (EDA) is carried out using correlation heatmaps, boxplots, and class distribution graphs
• Step 4: Base models including Random Forest (n=100) and Support Vector Machine (RBF kernel) are trained independently
• Step 5: A hybrid stacking approach is used where predictions from Random Forest and SVM are given as input to a Logistic Regression model acting as a meta-learner
• Step 6: The models are evaluated using accuracy, confusion matrix, classification report, and cross-validation
• Step 7: A Disease Risk Score ranging from 0 to 100 is calculated using model probabilities to indicate severity
• Step 8: SHAP explainability is used to analyze feature contributions for each prediction
• Step 9: The final model is used for prediction by loading the saved .pkl file, scaling new input data, and outputting plant health status along with confidence score

Chapter 4
Implementation

4.1 Methodology / Proposal

The project implements a complete machine learning pipeline across 9 Jupyter notebooks. Each notebook handles a distinct phase of the pipeline, making the code modular and reproducible.

4.1.1 Data Preprocessing

Timestamp and Plant_ID columns were dropped as they carry no predictive value. Labels were encoded using a manual mapping: Healthy=0, Moderate Stress=1, High Stress=2. Feature scaling was applied using StandardScaler (mean=0, std=1) to normalize the vastly different ranges of sensor readings. The dataset was split 80% training (960 samples) and 20% testing (240 samples) using stratified sampling.

4.1.2 Random Forest Model

Random Forest was configured with n_estimators=100, random_state=42, and n_jobs=-1 for parallel processing. The model was trained on the scaled training set and evaluated on the test set using accuracy score, classification report, and confusion matrix. Feature importance scores were extracted and visualized. Cross-validation (5-fold) confirmed generalization.

4.1.3 SVM Model

SVM was configured with kernel='rbf', C=1.0, gamma='scale', and probability=True (required for soft voting and stacking). The model was trained on the same scaled data and evaluated with identical metrics.

4.1.4 Hybrid Stacking Model (Novelty)

The hybrid model uses Scikit-learn's StackingClassifier with Random Forest and SVM as base estimators and Logistic Regression as the meta-learner. The meta-learner is trained on out-of-fold predictions from the base models using 5-fold cross-validation, preventing data leakage. This approach allows the meta-learner to learn the optimal weighting of RF and SVM predictions, achieving superior accuracy over either individual model.

4.1.5 Disease Risk Score

A continuous risk score (0-100) was derived from the model's probability outputs: Risk Score = (P(Moderate Stress) x 50) + (P(High Stress) x 100). This converts the probabilistic output into an actionable severity index. Score ranges: 0-30 (Safe), 31-60 (Monitor), 61-80 (Warning), 81-100 (Critical).

4.2 Testing / Verification Plan

Test ID	Test Case	Condition	System Behavior	Expected Result
T01	Healthy Prediction	Soil 60%, Humidity 45%, Chlorophyll 38	Model predicts Healthy	Healthy, >95% conf
T02	High Stress Prediction	Soil 15%, Humidity 88%, Chlorophyll 12	Model predicts High Stress	High Stress, >95% conf
T03	Risk Score	High Stress sample	Risk score computed	Score > 75
T04	Batch Prediction	2 samples input	Both predicted correctly	Healthy + High Stress
T05	Model Saving	After training	PKL files saved	Files exist in /models

4.3 Result Analysis

The following table summarizes the performance of all three models on the test set:

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	~96%	High	High	High
SVM (RBF)	~93%	High	Moderate	Moderate
Hybrid Stacking	~98%	Highest	Highest	Highest

Key findings from result analysis: The Hybrid Stacking model outperformed both individual models. Prediction confidence on new sensor readings reached 98.51% (Healthy) and 98.85% (High Stress) as verified in notebook 08_prediction.ipynb. Chlorophyll Content and Electrochemical Signal were identified as the strongest predictors of plant stress via feature importance analysis.

4.4 Quality Assurance

Quality assurance was maintained through: (1) 5-fold cross-validation to prevent overfitting, (2) stratified train-test split ensuring balanced class representation, (3) notebook-based reproducible code with fixed random seeds (random_state=42), (4) version-controlled codebase on GitHub, and (5) modular pipeline with independent notebooks enabling individual step verification.

Chapter 5
Standards Adopted

5.1 Design Standards

In this project, basic design guidelines were followed based on common machine learning practices. The workflow roughly follows the CRISP-DM process, which includes understanding the problem, exploring the data, preparing it, building models, evaluating results, and then using the model. Simple flowcharts were created to understand and explain how the system works.

5.2 Coding Standards

The following coding standards were adhered to throughout the project:

- **Python Enhancement Proposal** (PEP) 8 Python style guide for code formatting and naming conventions
- Descriptive variable names (e.g., X_train, y_test, hybrid_model) for readability
- Functions are modular — each notebook handles a single pipeline stage
- All code is commented with clear section headers
- Fixed random seeds (random_state=42) for reproducibility
- Warning suppression (warnings.filterwarnings) for clean output
- File paths handled with pathlib.Path for cross-platform compatibility

5.3 Testing Standards

Basic testing methods were used to check model performance. The model was validated using 5-fold cross-validation to make sure it performs well on different data splits. Standard evaluation metrics like accuracy, precision, recall, and F1-score were used to measure performance. Some simple test cases were checked manually to see if the results matched expectations.

Chapter 6
Conclusion and Future Scope

6.1 Conclusion

This project built a complete machine learning pipeline to predict plant diseases at an early stage using biosensor and soil data. A hybrid model combining Random Forest and SVM with a stacking method was used, and it achieved around 98% accuracy, performing better than using each model separately.

The project mainly focused on detecting plant stress before visible symptoms appear. It also introduced a few useful ideas, such as using a combined RF-SVM model, creating a Disease Risk Score (0–100) to show how serious the condition is, and using SHAP to understand which features influence the predictions. From the results, chlorophyll content and electrochemical signals were found to be the most important factors.

Compared to image-based methods, this approach is more practical because it does not require cameras and can continuously monitor plants using sensors.

6.2 Future Scope

There are several ways this project can be improved or extended in the future:

- Use more data with different plant types and regions to make the model more reliable
- Try advanced models like LSTM or Transformers to better understand time-based data
- Build a mobile app for farmers to get real-time alerts
- Connect the system with hardware like Raspberry Pi or Arduino for real-time field use
- Improve the model to identify specific diseases instead of just stress levels
- Explore federated learning so multiple farms can train models without sharing data directly

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